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ABC Logistic

Vehicle Routing Problem Final Project

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Abstract

This report models a Vehicle Routing Problem (VRP) in the logistics company ABC located in Kayseri. The project aims to minimize total transportation cost by detecting the most appropriate transportation routes with mathematical models on GAMS. At this point, the mathematical model is optimized for the real-life problem using parameters, decision variables, as well as constraints such as transfer balance, flow, capacity, number of vehicles, time and load. This approach contributes lower costs, shorter travel distances, less transportation duration and more environmentally sustainable.

Keywords

Vehicle routing problem, optimization, cost saving, time, sustainability

1. INTRODUCTION

1.1 DESCRIPTION OF THE FIRM

ABC is a well-established logistics company founded in 1992 and has been providing comprehensive logistics services to companies operating in various sectors since then. With its experience and innovative corporate identity gained in the sector since its establishment, ABC has positioned itself as a reliable and strategic business partner in the logistics sector. The company aims to continuously develop by integrating modern logistics applications and customer-focused solutions into its operations.

ABC's main areas of expertise are effective management of transportation, storage and distribution processes. It contributes to companies' efficient use of time and resources in their supply chain processes by offering strategic solutions specific to customer needs. Thanks to this customer-focused approach, ABC aims to create not only operational efficiency but also long-term value.

The company's headquarters is in Kayseri, one of Turkey's strategic regions. The facility, built on 5,000 m² of closed and 30,000 m² of open space, is the main center where the company conducts a significant portion of its domestic and international logistics operations.ABC stands out with its transportation activities worldwide, especially in Asia and Europe. It offers fast, reliable and flexible logistics solutions to its business partners by using different transportation modes such as land, air, sea and rail transportation in an integrated manner. Combining its global reach with its local expertise, ABC continues to respond effectively to the changing needs of its customers in a dynamic and competitive business environment.

1.2 CURRENT SYSTEM IN THE COMPANY

Direct (İ, K)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0

Table 1 is there a direct connection btw start nodes and intermediate nodes

Direct (K1, K2)	6	7	8	9	10	11	12	13	14	15	16	17	18
6	0	0	0	0	1	1	1	0	0	0	0	0	0
7	0	0	0	0	1	1	1	0	0	0	0	0	0
8	0	0	0	0	0	0	0	1	1	1	1	0	0
9	0	0	0	0	0	0	0	1	1	1	1	0	0
10	0	0	0	0	0	0	0	1	1	1	1	0	0
11	0	0	0	0	0	0	0	1	1	1	1	0	0
12	0	0	0	0	0	0	0	0	1	1	1	0	0
13	0	0	0	0	0	0	0	0	1	1	1	0	0
14	0	0	0	0	0	0	0	0	0	0	0	1	0
15	0	0	0	0	0	0	0	0	0	0	0	1	0
16	0	0	0	0	0	0	0	0	0	0	0	1	0
17	0	0	0	0	0	0	0	0	0	0	0	0	1
18	0	0	0	0	0	0	0	0	0	0	0	0	0
19	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 2 is there a direct connection btw intermediate node

Direct(K,J)	19	20	21	22	23	24	
6	0	0	0	0	0	0	
7	0	0	0	0	0	0	
8	0	0	0	0	0	0	
9	0	0	0	0	0	0	
10	0	0	0	0	0	0	
11	0	0	0	0	0	0	
12	0	0	0	0	0	0	
13	0	0	0	0	0	0	
14	1	1	1	1	1	1	
15	1	1	1	1	1	1	
16	1	1	1	1	1	1	
17	0	0	0	0	0	0	
18	0	1	0	0	0	0	

Table 3 is there a direct connection btw intermediate nodes and end nodes

S(i, K)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	0	0	0	0	0	1	1	0	0	1	1	1	0	0	0	0	0	0
2	0	0	0	0	0	1	1	0	0	1	1	1	0	0	0	0	0	0
3	0	0	0	0	0	1	1	0	0	1	1	1	0	0	0	0	0	0
4	0	0	0	0	0	1	1	0	0	1	1	1	0	0	0	0	0	0
5	0	0	0	0	0	1	1	0	0	1	1	1	0	0	0	0	0	0

Table 4 is there a passage btw start nodes and intermediate nodes by sea

S (K1, K2)	6	7	8	9	10	11	12	13	14	15	16	17	18
6	0	0	0	0	1	1	1	0	0	0	0	0	0
7	0	0	0	0	1	1	1	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0	0	0	0	0
15	0	0	0	0	0	0	0	0	0	0	0	0	0
16	0	0	0	0	0	0	0	0	0	0	0	0	0
17	0	0	0	0	0	0	0	0	0	0	0	0	1
18	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 5 is there a passage btw intermediate nodes by sea

Distance (İ, K)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	0	0	0	0	0	2025	1674	3323	3474	3700	3148	3289	0	0	0	0	0	0
2	0	0	0	0	0	1110	946	2408	2559	2780	2881	2222	0	0	0	0	0	0
3	0	0	0	0	0	950	984	1963	2115	5800	6000	7400	0	0	0	0	0	0
4	0	0	0	0	0	679	480	1977	1418	2040	2037	2537	0	0	0	0	0	0
5	0	0	0	0	0	1097	1608	1879	2030	4500	4460	4150	0	0	0	0	0	0

Table 6 is there a passage btw intermediate nodes and end nodes by sea

S(K, J)	19	20	21	22	23	24
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
11	0	0	0	0	0	0
12	0	0	0	0	0	0
13	0	0	0	0	0	0
14	0	0	0	0	0	0
15	0	0	0	0	0	0
16	0	0	0	0	0	0
17	0	0	0	0	0	0
18	0	0	0	0	0	0

Table 7 Distance btw starts nodes and intermediate nodes

Distance (K1, K2)	6	7	8	9	10	11	12	13	14	15	16	17	18
6	0	0	0	0	1670	1670	2400	0	0	0	0	0	0
7	0	0	0	0	2605	2600	2200	0	0	0	0	0	0
8	0	0	0	0	0	0	0	850	1300	1450	1300	0	0
9	0	0	0	0	0	0	0	800	1250	1400	1250	0	0
10	0	0	0	0	0	0	0	600	1100	1200	1100	0	0
11	0	0	0	0	0	0	0	550	1120	1220	1130	0	0
12	0	0	0	0	0	0	0	250	850	950	700	0	0
13	0	0	0	0	0	0	0	0	987	940	422	0	0
14	0	0	0	0	0	0	0	0	0	0	0	736	0
15	0	0	0	0	0	0	0	0	0	0	0	797	0
16	0	0	0	0	0	0	0	0	0	0	0	1719	0
17	0	0	0	0	0	0	0	0	0	0	0	0	286
18	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 8 Distance btw intermediate nodes

Distance (K, J)	19	20	21	22	23	24
6	0	0	0	0	0	0
7	0	0	0	0	0	0
8	0	0	0	0	0	0
9	0	0	0	0	0	0
10	0	0	0	0	0	0
11	0	0	0	0	0	0
12	0	0	0	0	0	0
13	0	0	0	0	0	0
14	551	1600	1103	252	1323	1718
15	684	1200	1000	540	1118	1369
16	1605	1800	362	1317	1048	2317
17	0	0	0	0	0	0
18	0	560	0	0	0	0

Table 9 Distance btw intermediate nodes and end nodes

ABC Logistics' transportation operations offer fast, reliable and flexible solutions in both Asian and European markets with the integrated use of land, sea modes. The company's service network includes various border gates, ports and distribution points spread across different countries and regions. Aktaş, Gürbulak and Habur are among the important border gates used by the company in road transportation, and access to countries such as Azerbaijan, Turkmenistan, Syria, Georgia, Iraq and Iran is provided through these gates. Road connections to Europe are carried out via the Bulgarian Kapıkule and Greek İpsala border gates. Land transportation from Turkey to countries such as Italy Trieste, Italy, Germany, France, Spain and Poland also has a significant share.

Pendik, Yalova and Mersin ports are used in sea transportation. Intensive sea transportation is carried out from European countries (Spain, France, Germany, Italy, Poland) to these ports. In addition, connections to Turkmenistan are provided via the Baku and Turkmenbashi ports.

The above tables and map visuals show the different routes used by ABC Logistics and their distances. Transportation costs and routes are optimized by taking into account parameters such as first-hand measured kilometer-based costs, transportation time and transportation capacity. In this way, ABC Logistics aims to provide more economical, faster and sustainable logistics solutions to its customers.

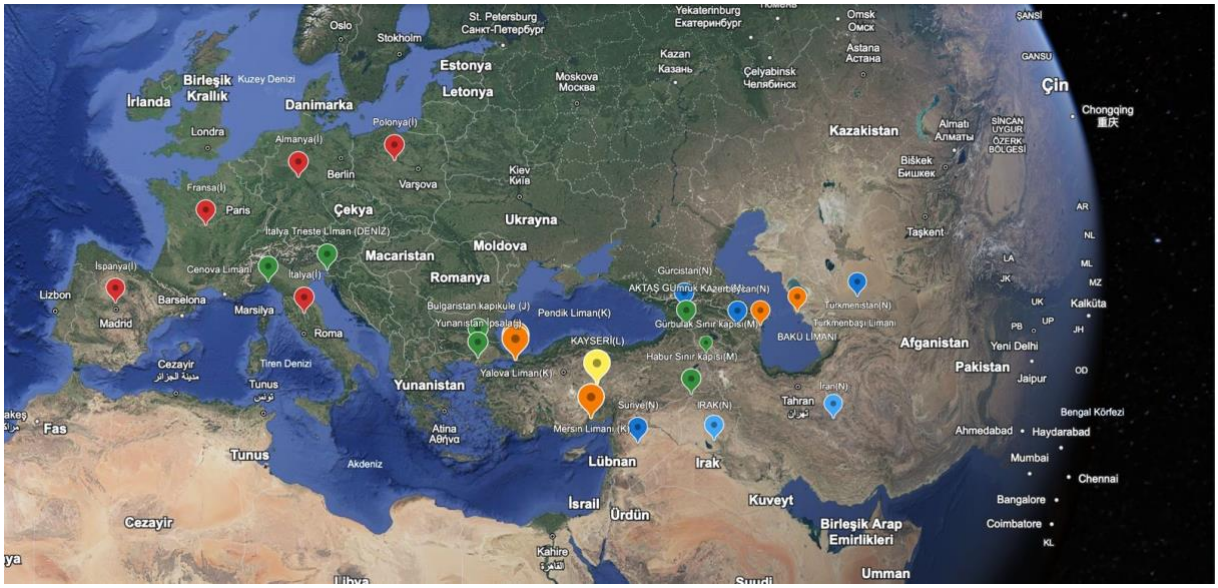

Figure 1 Routes

2. LITERATURE REVIEW

The vehicle routing problem (VRP) is an optimization problem which aims to service the customers from start to end point with the most efficient routes for a fleet of vehicles. With developing the transportation network all around the world, various sub-branches of VRP have been released especially such as multiple depots, multiple transportation modes, capacity constraints, and time windows are common features in modern logistics networks. To understand the real-world supply chain operations, these constraints play crucial role, additionally to optimizing route planning and minimizing costs with advanced solution techniques.

In modern logistics systems, not only road route planning but also the use of multiple transport modes such as sea and air is important to optimize route planning in long-distance international transport. Cattaruzza. (2016) introduced the multimodal vehicle routing problem (MMVRP) and developed models that require vehicles to plan routes with transitions between nodes, such as a truck dropping off a container at a ship in port. On the other hand, Jabali et al. (2012) states that using only short and low-cost routes will not be enough to reduce emissions, and also environmental and sustainability in the transportation sector can be achieved with electric vehicles.

Various solution technics have been developed for the complexity and NP-hard nature of VRP. Metaheuristics provides efficient solutions for large-scale VRPs. These algorithms are classified as genetic algorithms, tabu search, and ant colony optimization (ACO). Gendreau et al. (2001) achieved very good results in the multiple depots problem using the tabu search method. Nowadays, with the development and widespread use of machine learning and deep learning, great results can be obtained with deep learning and reinforcement learning for these complex problems. Mousavi et al. (2020) optimized the deep reinforcement learning method for the vehicle routing problem, allowing vehicles to drive autonomously along the route. In this way, improvements in time and fuel are achieved.

3. PROBLEM DEFINITION

In modern logistics, companies are challenged with optimizing their transportation networks. These optimizations require a balanced cycle between costs, customer expectations and satisfaction, and operational efficiency. However, the logistics process itself is not always balanced. There are constant instantaneous developments, such as customer demand changes, roadworks, and more. Even if everything is perfectly balanced, this is not enough, because a rival company can get ahead of you with better optimization.

The fundamental target of this project is to determine the most appropriate transportation routes that will minimize the total transportation cost, considering different transportation nodes, ports, border gates and loading points. At the same time, waiting times at ports, transportation capacities and demands of countries are also considered in the model; the problem is handled as a deterministic optimization model.

4.SOLUTION METHODOLOGY

The Vehicle Routing Problem (VRP) is a well-known problem in operational research where customers of known demands are supplied by one or several depots.

ABC company is a logistics company that transports goods from European countries to Asian countries. It transports goods from many European countries. But the leading countries are Spain, France, Germany, Italy and Poland.

Sets

i = 1,2, ..., 5

k = 6,7, ..., 18

j = 19,20, ..., 24

k1, k2: Aliases of k. There are used for multi-level transfers

Parameters

$a_{ik} \in \{0,1\}$ Is it possible to reach point from point i to j?

$direct_{ik} \in \{0,1\}$ Is it possible to go directly from point i to point j?

$s_{ik} \in \{0,1\}$ Is it possible to cross by sea?

$distance_{ik}$: distance i to j

$travel_{ik}$: travel duration i to j

p^s: sea transport cost / km

p^r: road transport cost / km

T^s : sea transport cost / tones

T^r : road transport cost / tones

Q : load capacity of a truck

V : total truck number

$supply_i$: Supply quantity at node i

$demand_j$: Demand quantity at node j

T_{max}: maximum total transportation time (21)

Decision Variables

Binary Variables:

X_{ik} : Is there a transition from i to k?

Y_{kk1} : Is there a transition from k to k1?

W_{kj} : Is there a transition from k to j?

Positive Variables:

$m1_{ik}$: Amount shipped from i to k

$m2_{kk1}$: Amount shipped from k to k1

$m3_{kj}$: Amount shipped from k to j

z : Total cost (objective function)

Objective Function

Minimize total cost (z):

$$z = \sum_i \sum_k \left[\left(p^s * s_{ik} + p^r * (1 - s_{ik})\right) * distance_{ik} * X_{ik} + \left(T^s * s_{ik} + T^r * (1 - s_{ik})\right) * m1_{ik}\right] +$$

$$\sum_k \sum_{k1} \left[\left(p^s * s_{kk1} + p^r * (1 - s_{kk1})\right) * distance_{kk1} * Y_{kk1} + \left(T^s * s_{kk1} + T^r * (1 - s_{kk1})\right) * m2_{kk1}\right] +$$

$$\sum_{k1} \sum_j \left[\left(p^s * s_{k1j} + p^r * \left(1 - s_{k1j}\right)\right) * distance_{k1j} * W_{k1j} + \left(T^s * s_{k1j} + T^r * \left(1 - s_{k1j}\right)\right) * m3_{k1j}\right]$$

Constraints

1. Supply balance (at supplier nodes i):

$$\sum_k a_{ik} * m1_{ik} = supply_i \qquad \forall i$$

2. Intermediate flow balance at first level (nodes k):

$$\sum_i a_{ik} * m1_{ik} = \sum_k abtw_{kk1} * m2_{kk1} \qquad \forall k$$

3. Intermediate flow balance at second level (nodes k1):

$$\sum_k abtw_{kk1} * m2_{kk1} = \sum_j a2_{k1j} * m3_{k1j} \qquad \forall j$$

4. Demand satisfaction:

$$\sum_k a2_{kj} * m3_{kj} \geq demand_j \qquad \forall j$$

5. Flow balance at middle transfer node k1:

$\sum_k Y_{kk1} = \sum_k Y_{k1k}$ $\forall k1$

(with both incoming and outgoing arcs)

6. Capacity constraints:

$m1_{ik} \leq Q * X_{ik}$ $\forall ik$

$m2_{kk1} \leq Q * Y_{kk1}$ $\forall kk1$

$m3_{kj} \leq Q * W_{kj}$ $\forall kj$

7. Route validity constraints:

$X_{ik} \leq a_{ik} * direct_{ik}$ $\forall ik$

$Y_{kk1} \leq abtw_{kk1} * directbtw_{kk1}$ $\forall kk1$

$W_{kj} \leq a2_{kj} * direct2_{kj}$ $\forall kj$

8. Flow allowed only on valid arcs:

$m1_{ik} \leq Q * a_{ik} * direct_{ik}$ $\forall ik$

$m2_{kk1} \leq Q * abtw_{kk1} * directbtw_{kk1}$ $\forall kk1$

$m3_{kj} \leq Q * a2_{kj} * direct2_{kj}$ $\forall kj$

9. Vehicle limit:

$\sum_i \sum_k X_{ik} + \sum_k \sum_{k1} Y_{kk1} + \sum_k \sum_j W_{kj} \leq V$

10. Time constraint:

$\sum_i \sum_k travel_{ik} * X_{ik} + \sum_k \sum_{k1} travelbtw_{kk1} * Y_{kk1} + \sum_k \sum_j travel2_{kj} * W_{kj} \leq T_{max}$

5.SENSITIVITY ANALYSIS

Total Cost vs Vehicle Capacity (Q)

The shaded area on the graph shows the variability, i.e. the cost range according to the scenarios. This range narrows slightly as the capacity increases, i.e. the cost estimate seems more stable at high capacities.

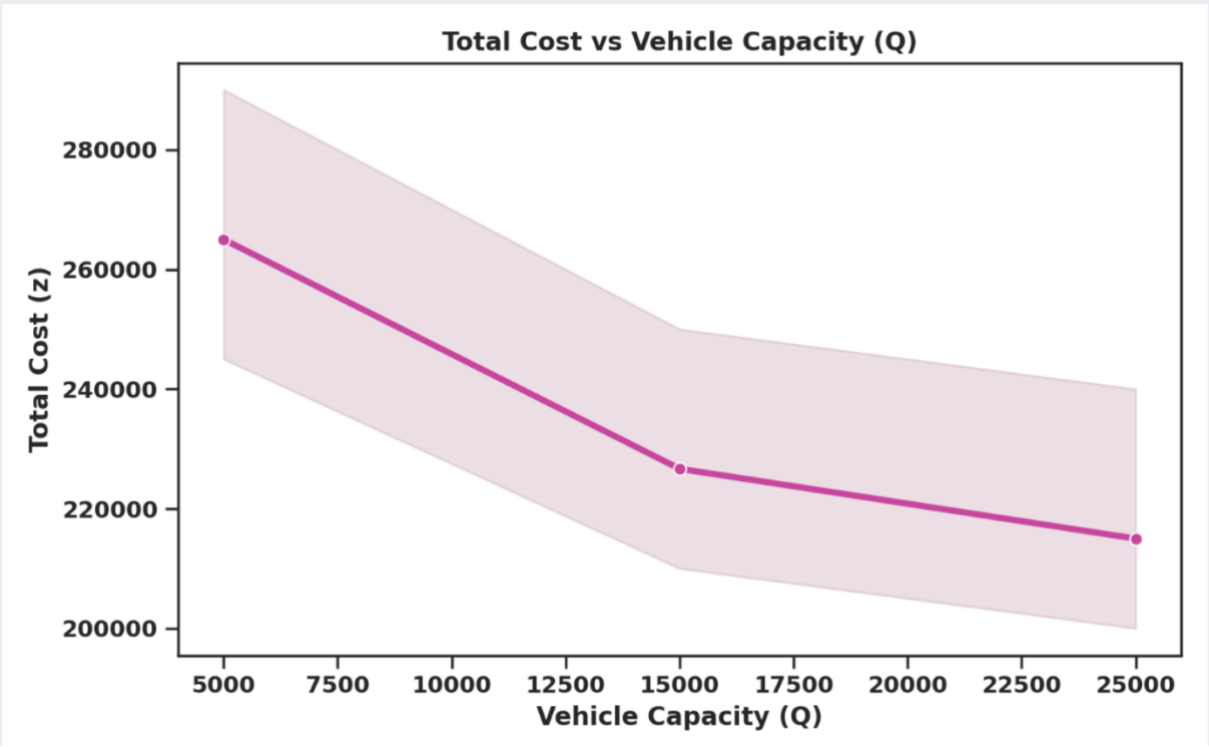

Figure 2 Total Cost vs Vehicle Capacity (Q)

Total Cost vs Tmax (Maximum Travel Time) for different Q values

Here, the total cost is shown against the maximum travel time limit for different vehicle capacities. The graphs show a decreasing trend for all Q values, i.e. the total cost decreases as Tmax increases. At this point, we can comment that as the maximum travel time increases, there is more flexibility in route selection. Perhaps longer but more cost-effective routes can be chosen. This reduces the total cost. Q=5000 (blue line) is in the highest cost range, while Q=25000 (green line) is in the lowest cost range. This shows the dominant effect of vehicle capacity on cost and that Tmax provides significant flexibility to optimize cost.

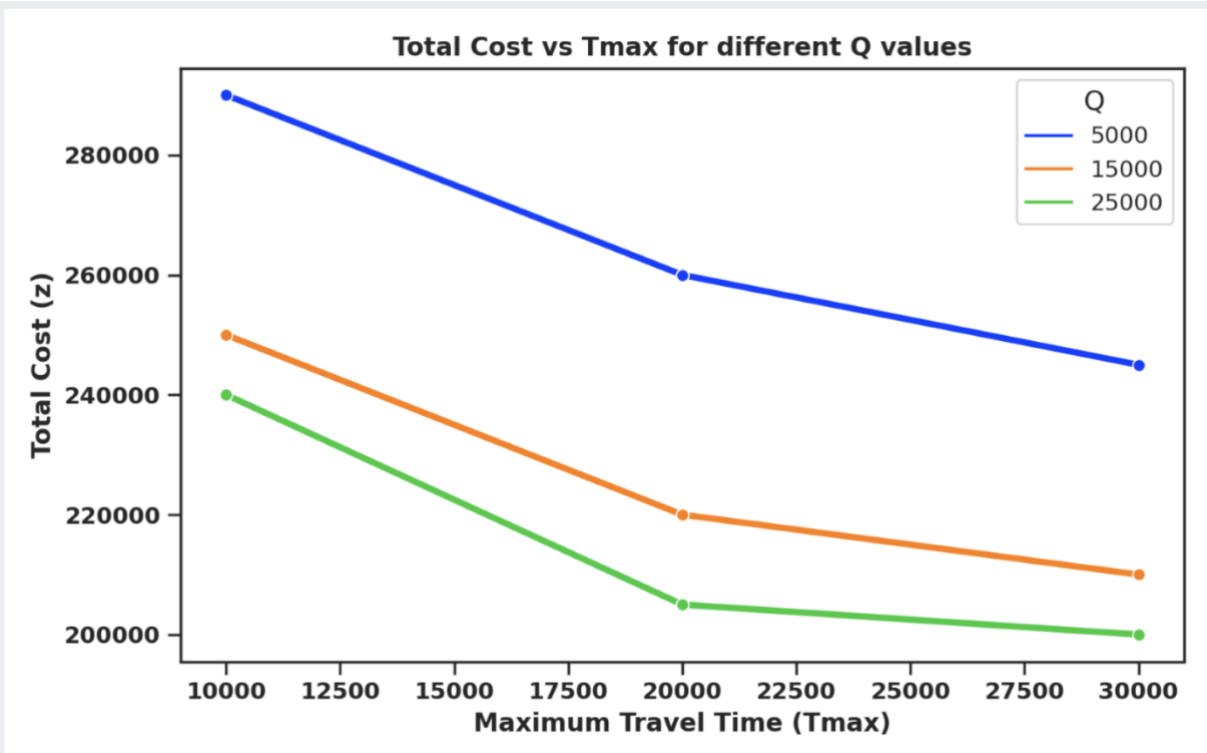

Figure 3 Total Cost vs Tmax for different Q values

Total Cost vs Number of Vehicles (V) for different Q

In this graph, it is observed that the total cost decreases as the number of vehicles increases. As the number of vehicles increases, the load per vehicle decreases and thus the need for high-capacity decreases. However, very high vehicle numbers can be costly in practice in terms of logistics and operations. Therefore, using or not using a vehicle capacity constraint in modeling is important for optimization.

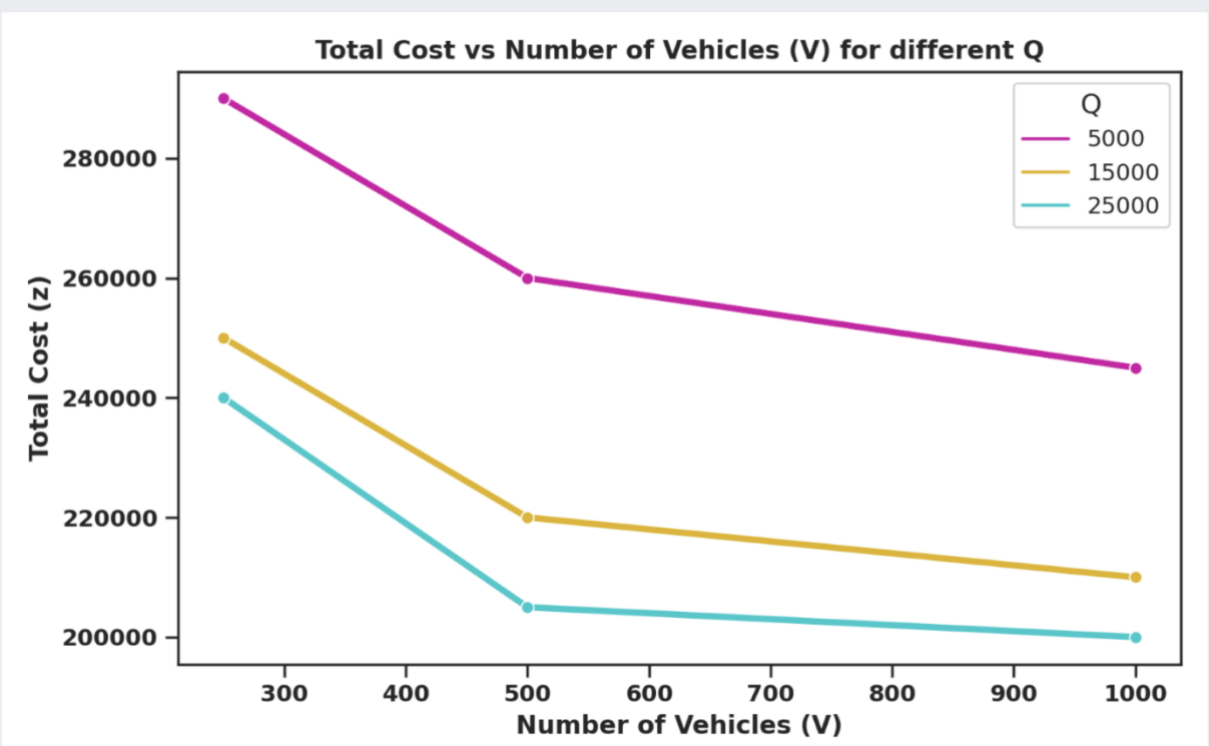

Figure 4 Total Cost vs Number of Vehicles (V) for different Q

6.RESULTS

In this study, the Multimodal Transportation Model developed for Kayseri-based logistics company ABC aims to minimize total transportation costs by integrating different transportation modes (road and sea). The model was optimized in the GAMS program using real-world data and parameters.

The results show that thanks to the use of multimodal transportation;

Total transportation costs have decreased significantly. The difference between road and sea costs has been effectively evaluated by the model, and the most cost-effective routes have been preferred.

The balance of demand and supply has been maintained. In the models, demand amounts on a country basis have been fully met, and the continuity of logistics operations has been secured.

The flexible decision to stop at intermediate transfer points (e.g. Kayseri) has increased operational efficiency and made a positive contribution to route optimization.

The waiting time constraint at ports has been successfully implemented, minimizing delays caused by waiting.

The model has produced realistic solutions considering the number of trucks in the fleet and capacity constraints.

The best optimal solution, showing total cost for 600 trucks, was obtained on GAMS as 4012985.2500.

Operational time efficiency has been optimized by considering transportation time constraints.

The model increased the resilience of the logistics network by using alternative routes and modes. It offers flexible routes, especially against possible blockages at border gates or port delays. Detailed examination of fixed and variable costs allowed the identification of the highest cost items, which allowed the development of cost-cutting strategies.

In addition, sensitivity analyses of parameters such as p^s, p^r, T^s, T^r used in the model were performed and it was observed that the cost advantages changed in line with the differences between transportation modes.

Best possible	4012985.2500
Absolute gap	0
Relative gap	0
Execution time	0.344 seconds

7.CONCLUSION

The mathematical model developed within the scope of this project provided optimization in terms of both cost and time in the transportation operations of ABC logistics company from Europe to Asia. The multimodal transportation approach determined the most suitable routes by considering the cost and time differences of land and sea transportation.

With the implementation of the model; Significant cost savings were provided to the company. Continuity and reliability in transportation processes increased. Waiting times at ports and border gates were taken under control and customer satisfaction was increased. The transportation capacity of the fleet was used effectively and logistics resources were managed at an optimum level. This model can be used as a decision support tool in the strategic planning and daily operations of ABC logistics company. It can play a critical role in creating competitive advantage, especially by optimizing the fleet and transportation infrastructure.

As a result, the optimization model presented by the study has become a powerful tool for the sustainable, fast and economical management of logistics processes. The model is of a nature that will guide ABC company's operational decisions in practice, can be expanded with different parameters and can produce advanced solutions by being updated with real-time data.

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